

An update rule for **[Gradient Descent]**, defined as

$$\begin{aligned}\Delta w^{(\tau)} &\leftarrow \alpha \Delta w^{(\tau-1)} - \epsilon \nabla_{\theta} \left(\frac{1}{m} \sum_{i=1}^m L \left(f(x^{(i)}; \theta), y^{(i)} \right) \right) \\ \theta &\leftarrow \theta + \Delta w\end{aligned}$$